Functional Design Document — Manufacturing Line (CPG)

Purpose

This Functional Design Document (FDD) defines the functional requirements, architecture, interfaces, control logic, safety considerations, testing and deployment plan for a manufacturing line in a Consumer Packaged Goods (CPG) facility. It is intended for automation engineers, manufacturing engineers, operations, QA, IT (MES/ERP) and stakeholders responsible for design, integration, commissioning and maintenance.

How to use this template

- Fill plant- and line-specific information in each section. Replace placeholders with diagrams, I/O tables and interface specs.
- Attach detailed PLC I/O tables, MES interface schemas and FAT/SAT scripts as appendices or separate files and reference them here.
- Use the checklists before FAT and commissioning to ensure completeness.

Document metadata

Version: 0.1
Author:
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Related documents: P&ID, Electrical Single Line, Standards, Safety regs

Executive summary

Provide a short description of the line (product family, packaging, target throughput) and the primary business goals (OEE, throughput, yield, changeover time).

Scope

extbfIn scope: e.g. filling, capping, labeling, inspection, case packing, palletizing.

extbfOut of scope: e.g. utilities, warehouse systems unless explicitly stated.

extbfAssumptions: e.g. existing MES available, plant network in place, spare parts available.

Stakeholders & roles

Business owner / Product owner; Manufacturing engineer; Automation engineer (PLC/HMI); Integration / MES engineer; QA; IT/Network; Safety/EHS; Maintenance.

Background & context

Link to floorplan and existing automation. Note constraints such as space, regulatory, or legacy systems to integrate.

Goals and success criteria

List measurable goals and acceptance criteria. Example:

- Throughput: $\geq$ 1200 units/hour measured over an 8-hour shift.
- OEE target: $\geq$ 85% after 90 days.
- Quality: critical defect rate $<$ 0.5%.

System overview / architecture

Describe the major subsystems and their responsibilities:

- Field layer: sensors, actuators, drives, safety relays.
- Control layer: PLC(s), safety PLC, HMI/operator panels.
- Supervisory: SCADA, Historian.
- Execution: MES (orders, genealogy, traceability).
- Enterprise: ERP.
- Network and cybersecurity zone segmentation.

extbfDiagram placeholders:

- *Topology diagram:* insert as a figure (e.g., `/diagrams/network_topology.png`) or embed a vector image.
- *PLC rack/layout:* attach a PDF or PNG of rack layout.

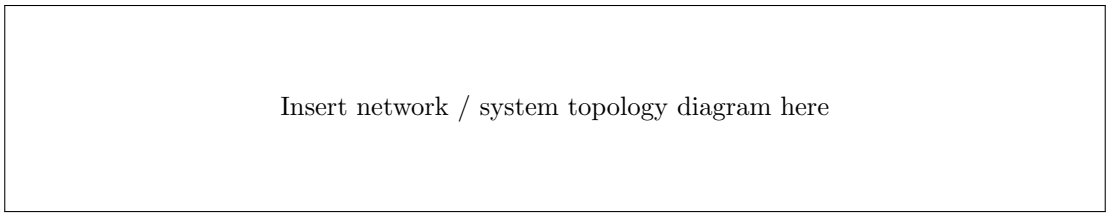


Figure 1: System topology (placeholder)

Functional components and responsibilities

For each subsystem include purpose, inputs, outputs, and acceptance criteria. Example entries:

Conveyors: Purpose – product movement and buffering. Inputs – start/stop, speed setpoint, sensors. Outputs – encoder feedback, fault signals. Acceptance – belt speed within ±5% under load.

Filling station: Purpose – dispense target volume with accuracy. Inputs – recipe, product ID, level sensors. Outputs – valve commands, dispense logs. Control – closed-loop dosing.

Capping: Purpose – apply/torque caps. Include torque checks and presence sensors.

Control strategies and sequences

Describe PLC state machines and key sequences: start/stop, emergency stop, changeover, recipe download and validation, manual/auto modes.

extbfExample (sequence):

1. Operator initiates `LineStart`; PLC verifies interlocks and safety.

2. PLC starts conveyors and brings machines to ready state.
3. PLC begins production on first product after recipe verification.
4. On fault, PLC transitions to `Fault`; operator resets and returns to `Idle` or `Running` depending on checks.

Data model and information flows

List events and payloads sent to MES/ERP/SCADA. Example fields and a simple table:

Field	Type	Descri
productionOrderId	string	MES
productCode	string	SKU
timestamp	ISO8601	event
deviceId	string	PLC
status	enum	start

Specify data retention policy (e.g., 1s raw data for 90 days, aggregated metrics for 3 years).

Interfaces and integration

Provide an interface table for each integration pair. Template:

Interface	Protocol	Payload / Format
PLC – SCADA	OPC UA / Modbus	Tag list, alarms, co
SCADA – MES	REST / MQ	Production events,
MES – ERP	REST / EDI	Material issue, conf

PLC & I/O mapping

Summarize PLC hardware and provide or link to a detailed I/O matrix (CSV or appendix). Example CSV filename: `docs/plc_io.csv`.

Safety and EHS requirements

List safety devices (light curtains, E-stops, safety PLC), how safety interacts with control logic (safe stop vs immediate stop), interlocks and LOTO procedures, and applicable standards (OSHA, ISO 13849/IEC 61508/62061 as applicable).

Alarms & HMI requirements

Define alarm severities, example alarm table, and HMI screens required: Overview, Alarms, Recipes, Maintenance, Diagnostics.

Quality and inspection

Define inspection points, vision checks, sampling plans and gating rules (e.g., reject a run after X consecutive fails). State what inspection data is logged to MES/historian.

Testing & validation plan

extbfUnit tests: function block validation and edge cases.
newline extbfFAT: functional sequences, safety tests, integration tests and performance.
newline extbfSAT: site verification, environmental checks, full integration with MES/ERP.
newline Include acceptance sign-off signature block for stakeholders.

Deployment & commissioning

Describe FAT lab requirements, rollout phases, expected downtime, rollback plans and version control practice for PLC/HMI source. Recommend storing PLC/HMI source in a VCS with tags for FAT/SAT/PROD.

Monitoring, telemetry & maintenance

KPIs to monitor: OEE, line speed, downtime, MTBF. Define dashboard needs, alerting thresholds, and routine maintenance intervals and spare parts list.

Security

State network segmentation, firewall rules, user authentication for HMI/SCADA/MES, and secure update procedures for PLC/HMI firmware and application code.

Reliability, availability & scalability

Availability targets and redundancy options (redundant PLC, dual network paths). List failure modes and mitigations.

Risk assessment & mitigations

Enumerate major risks (single vendor, supply chain, single point of failure) and proposed mitigations.

Open issues / decisions required

List decisions that need to be made (PLC vendor, MES vendor/integration pattern, mechanical layout sign-off) and owners.

Appendix & artifacts

Reference diagrams and files to attach:

- Floorplan: `diagrams/floorplan.png`
- Network topology: `diagrams/network_topology.png`
- PLC rack layout: `diagrams/plc_rack.pdf`
- I/O matrix: `docs/plc_io.csv`

Fill-in checklist

Complete before FAT:

- ☐ PLC I/O table attached
- ☐ MES interface spec attached
- ☐ Safety plan reviewed and signed
- ☐ HMI screens mock-ups provided